

SEOJIN KANG

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RESEARCH INTERESTS

Time Series Analysis, Data Mining, On-Device AI

EDUCATION

B.E. in Computer Engineering

Sungkyul University, Anyang, Korea

Mar 2023 — Feb 2027 (Expected)

GPA: 3.68/4.5

CONFERENCES

[C.1] Improving CLIP-Based Object Retrieval Through Background Removal with SAM Segmentation

Aug 2025

PlatCon 2025 International Workshop on Computer Image and its Applications (CIA 2025)

Jeju, Korea

- [Seojin Kang](#), Donghyun Park, Youngmyoung Kang*

PROJECTS

[P.6] Low-cost, Low-power Road Icing Detection System (Graduation Project)

Sep 2025 — Present

Team Leader, System Design, AI

- [Seojin Kang](#), Jaeho Lee, Kyeongjun Park, Donghyun Park
- Supervisor: Youngmyoung Kang (Department of Computer Engineering, Sungkyul University)
- Leading the development of an efficient edge-based system for road icing detection utilizing LoRa communication, responsible for the AI model and overall system design.

[P.5] AI Image Studio: Context-Preserving Multi-Style Image Transformation System 🌐

Nov 2025 — Dec 2025

Personal Project

- 1st Place in the Computer Engineering Department Artificial Intelligence API Utilization Competition
- To overcome the compositional limitations of text-based generative models, developed a web platform that preserves the position and context of user-placed objects while transforming their style. Built an intuitive compositional design interface using Fabric.js and a real-time object background removal system via RemBG. Through the FLUX Dev model and prompt engineering, implemented an animation style transformation pipeline for eight styles: Studio Ghibli, Pixar, Frozen, Snow White, Cinderella, Beauty and the Beast, Aladdin, Retro.

[P.4] SKU Course Scheduler 🌐

Aug 2025 — Dec 2025

Personal Project

- To support college students' course registration planning, which requires consideration of complex constraints, developed an automated timetable generation system based on AI optimization algorithms. Applied Genetic Algorithms(GA), Simulated Annealing(SA), and a hybrid optimization method combining both approaches. This enabled the design of four optimization types based on user preferences (avoiding morning classes, securing lunch breaks, maximizing free periods, and evenly distributing classes by day) along with essential/exclusion course settings, achieving personalized scheduling. It was built using actual lecture data from Sungkyul University (24 departments, all lectures except a few).

[P.3] Refrigerator Management & Recipe Recommendation System 🌐

May 2025 — Jun 2025

Full-Stack Developer, System Design

- Jaeho Lee*, [Seojin Kang](#)*, Junhyun Sim, Habeen Jang, Taeyeong Han
- The Best Project Selection in the Software Engineering Course
- Built an image analysis engine based on VLM (Gemini-2.5-Flash) to implement a system that automatically recognizes and classifies ingredients inside refrigerators into four categories: frozen, refrigerated, vegetables, and room temperature. By analyzing stored ingredients, it proposes customized recipes across five categories (Korean, Chinese, Japanese, Western, and others) and implements smart purchase recommendations for missing ingredients. Using the Flask framework and session-based data management, completed a full-stack architecture enabling ingredient addition, automatic consumption during cooking, and real-time inventory tracking.

[P.2] LLM-powered Voice Morning Briefing Assistant

Apr 2025

Team Leader

- Seojin Kang*, Donghyun Park
- 6th Place in the Computer Engineering Department Artificial Intelligence API Utilization Competition
- Developed a Raspberry Pi 5-based assistant system that delivers personalized information via voice every morning for busy modern individuals. By integrating with the Public Data Portal (weather) and Naver News API, the system collects information summarized by an LLM (Gemma-2b) and delivers a natural voice briefing through TTS.

[P.1] Design of a Baby Cry Prediction System

Sep 2023

Team Leader, System Design

- Seojin Kang, Jinkyung Kang, Soyeon Sung, Jinwoong Jang
- 4th Place in the Computer Engineering Department Design Project Competition
- To address the difficulty novice parents face in interpreting their baby's cries, designed an AI prediction system that analyzes the cause of crying and provides real-time alerts. Classified infant crying sounds into five types—hunger, sleepiness, false crying, discomfort, and pain—aiming to build customized prediction models for each cause.

EXTRACURRICULAR ACTIVITIES

Academic Activities

International Conference on Information Technology and Computer Communications

Aug 2025

Participant

Yokohama, Japan

Training Programs

Global Lounge English Extracurricular Program

Sep 2025 — Dec 2025

Supervisor: CHAPMAN KEVIN WAYNE (Paideia College, Sungkyul University)

Big Data & AI/Deep Learning

Jan 2024

Supervisor: Kunyoung Son (Pricing Data Platform, Coupang)

Deep Learning Fundamentals

Jul 2023

Supervisor: Sangsoon Lim (College of Art and Technology, Chung-Ang University)

Leadership & Service

Yongin City Youth Policy Network

2026 — Present

Member, 취창업분과

Computer Engineering Soccer Club

2023 — Present

Team Captain

2024 — Present

HONORS & AWARDS & SCHOLARSHIPS

Honors & Awards

1st Place, Computer Engineering Department AI API Utilization Competition

Dec 2025

Project: [P.5] AI Image Studio: Context-Preserving Multi-Style Image Transformation System

6th Place, Computer Engineering Department AI API Utilization Competition

Jun 2025

Project: [P.2] Raspberry Pi 5-based LLM-powered Voice Morning Briefing Assistant

4th Place, Computer Engineering Department Design Project Competition

Nov 2023

Project: [P.1] Design of a Baby Cry Prediction System

Scholarships

Academic Excellence Scholarship (D-level)

Spring 2026

GyeongWon Scholarship

2025

Academic Excellence Scholarship (D-level)

Fall 2023